NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83536/21
27 March 1992

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 3PDT, LOW LEVEL TO 10 AMPERES, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED, ALL WELDED, DC COILS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-83536.

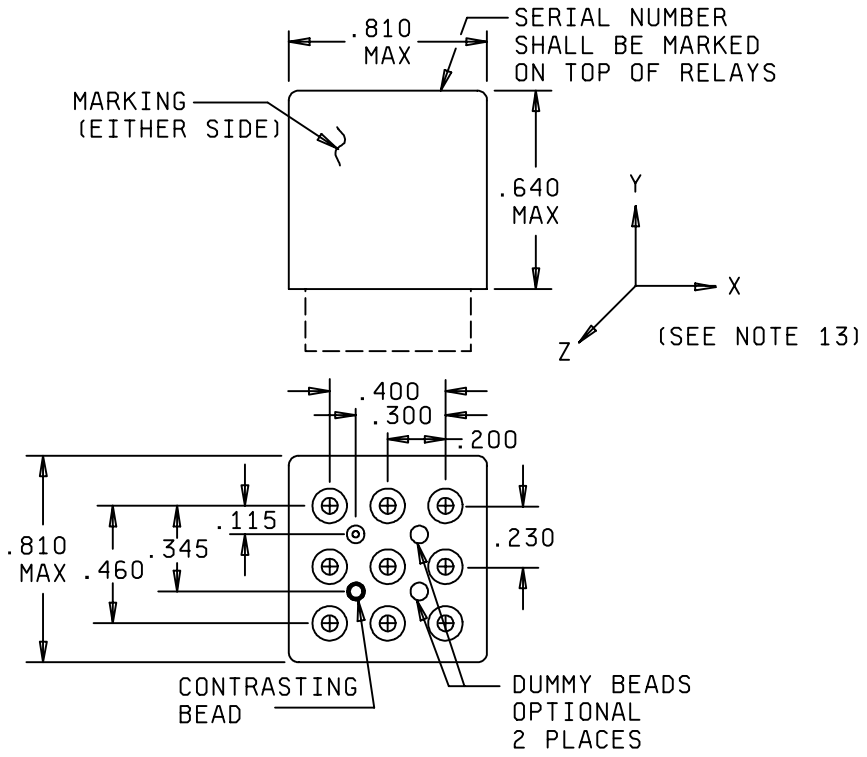
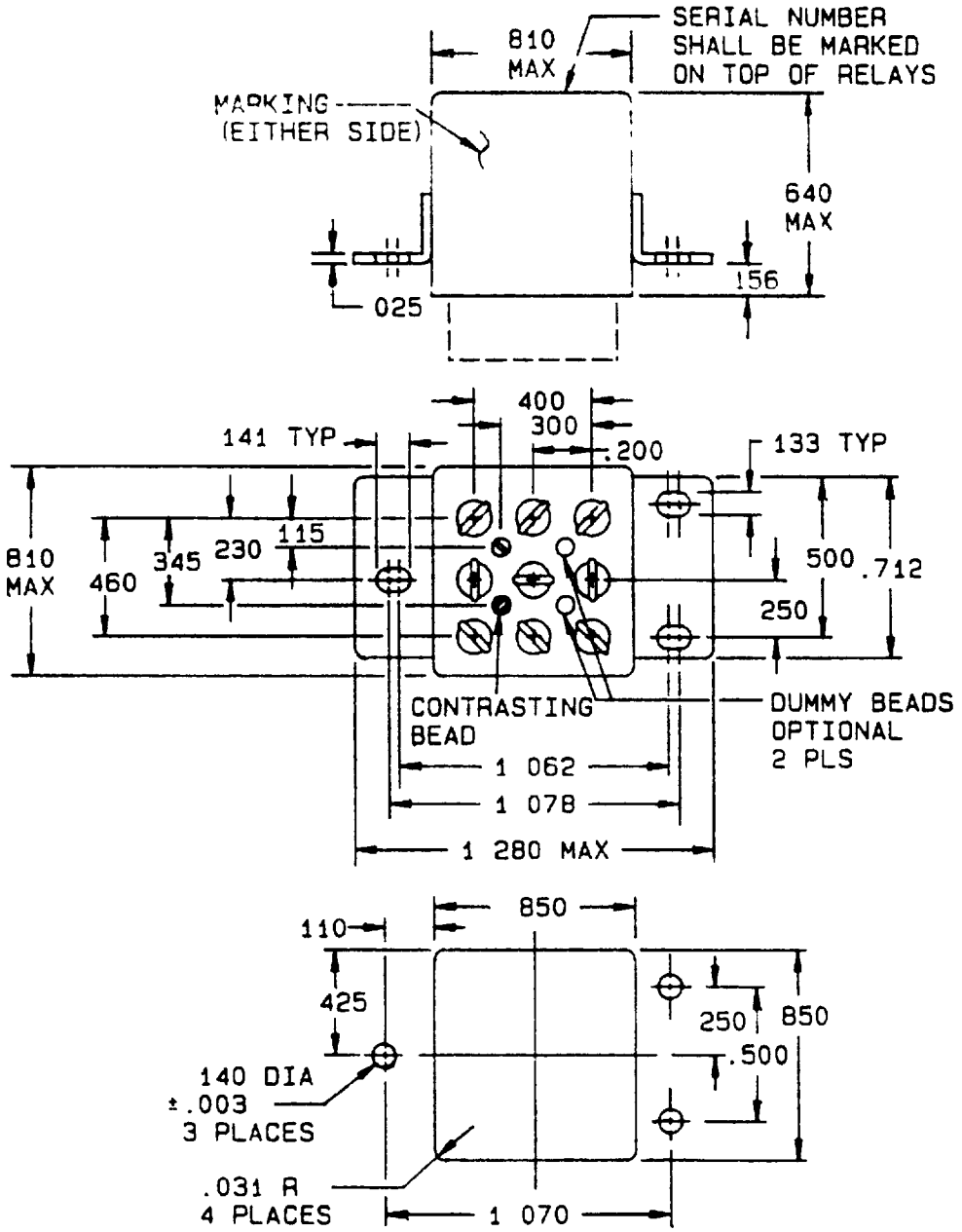


FIGURE 1. Dimensions and configurations.

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MOUNTING HOLE LAYOUT

RAISED VETRICAL FLANGE MOUNT (RVFM)

FIGURE 1 Dimensions and configurations - Continued

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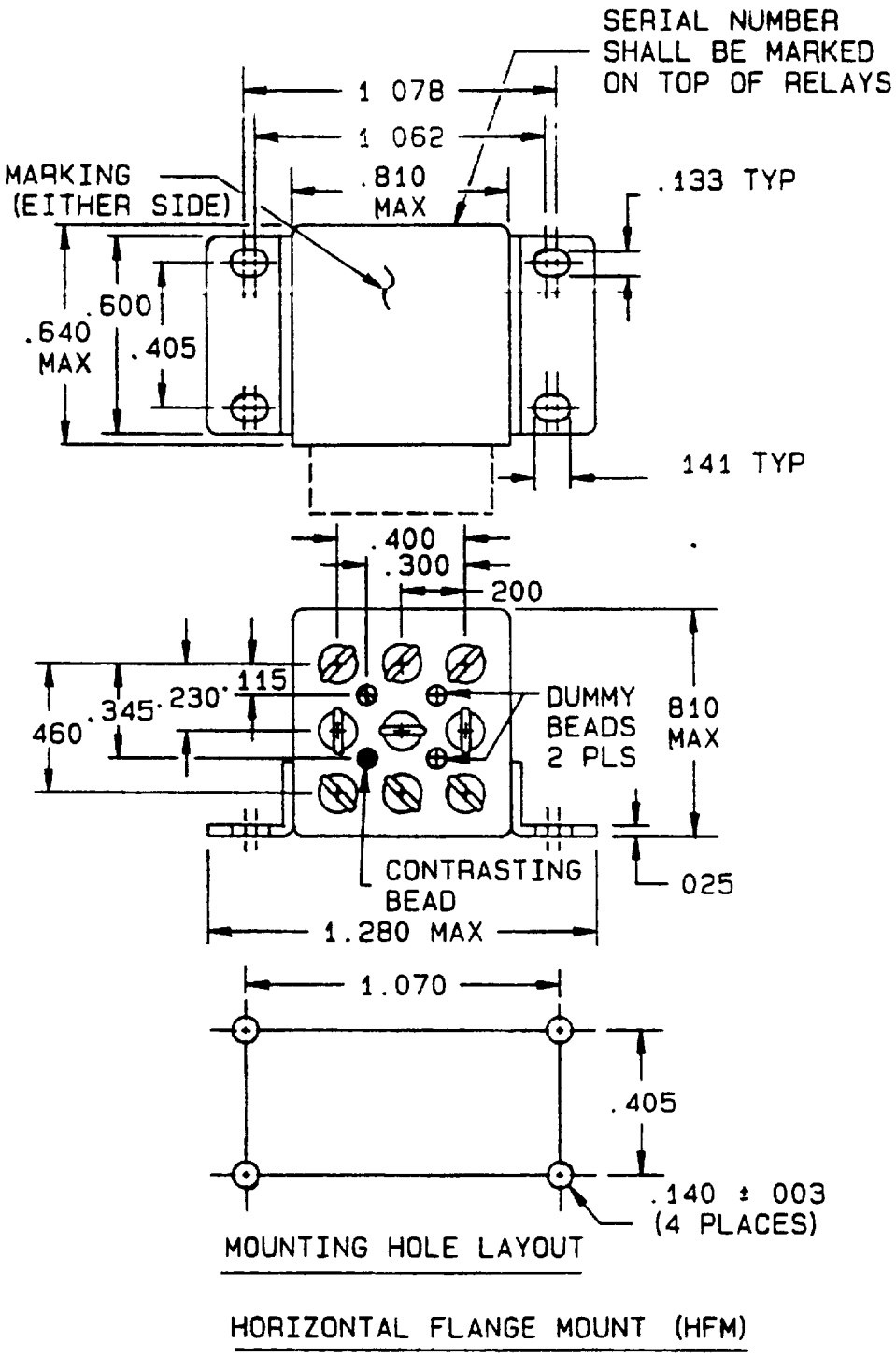


FIGURE 1 Dimensions and configurations - Continued

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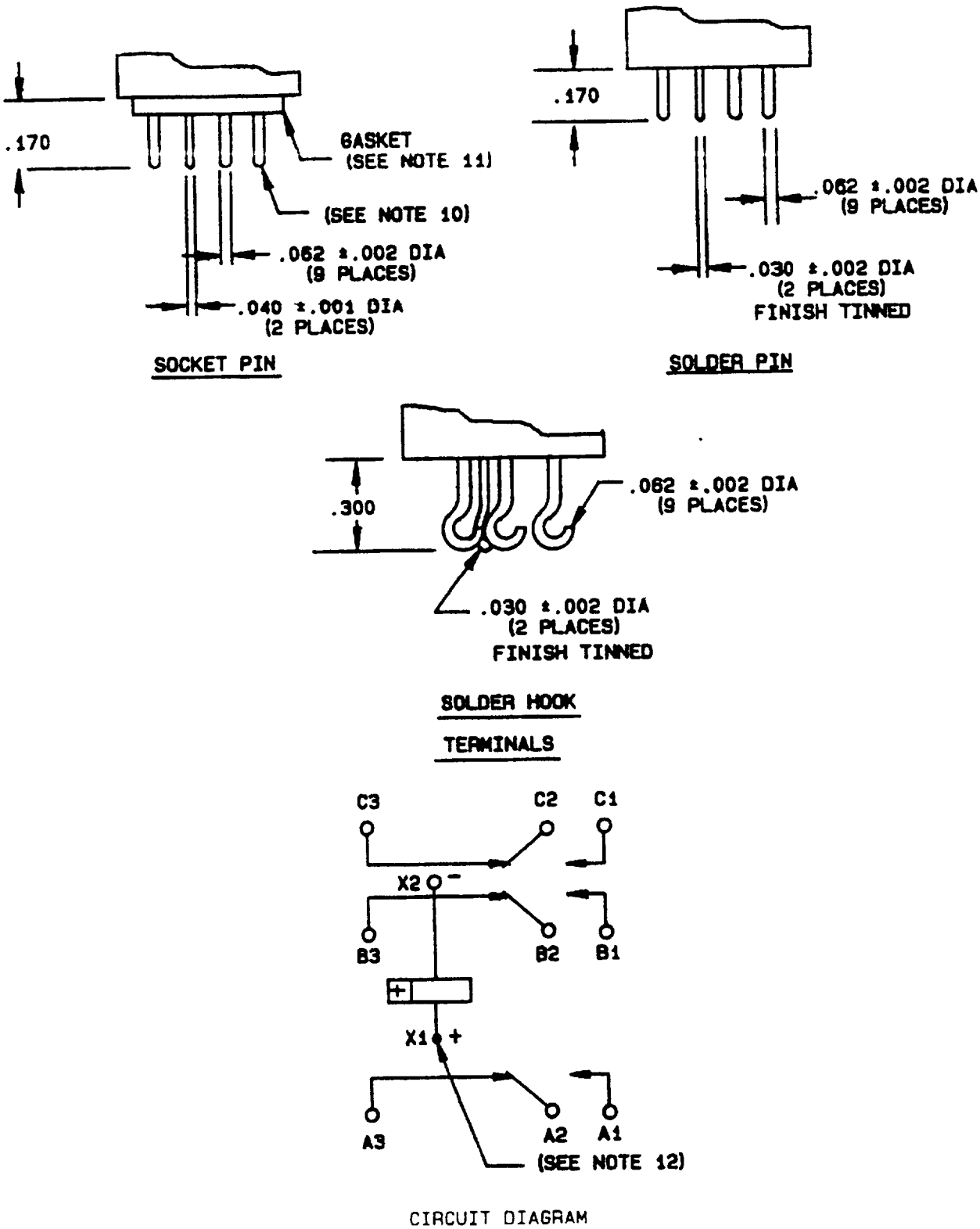


FIGURE 1 Dimensions and configurations - Continued

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Inches	mm	Inches	mm	Inches	mm
001	0 03	147	3 53	460	11 68
002	0 05	156	3 98	500	12 70
003	0 08	170	4 32	600	15 24
025	0 64	200	5 08	640	16 26
030	0 76	230	5 84	712	18 08
031	0 79	250	6 35	810	20 57
040	1 02	300	7 62	850	21 59
062	1.57	345	8 76	1 062	26 97
110	2 79	400	10 16	1 070	27 18
115	2 92	405	10 29	1 078	27 38
133	3 38	425	10 80	1 280	32 51
140	3 56				

NOTES

1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ± 010 (0 25 mm)
4 There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified
5 These relays are polarized monostable
6 This relay shall not operate or be damaged by reverse polarity
7 Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electro-magnet flux
8 Applicable to horizontal flange mount only The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side The remaining portion of the nameplate data shall be marked on the far side
9 Applicable to horizontal flange mount only Relays shall be marked with the manufacturer's name or CAGE code and date code Marking shall be with the bottom of the print adjacent to the near side
10 Socket pin terminals shall be gold-plated in accordance with MIL-G-45204, type II, class I The nickel underplating shall be 50 to 150 microinches thick minimum
11 Silicone rubber gasket, AMS 3332, shore hardness 15 to 35, thickness 050 ± 005 (1 27 ±0 13 mm)
12 Indicated terminal shall be identified by a contrasting bead
13 Plane of critical motion for vibration and shock is Y-axis

FIGURE 1 Dimensions and configurations - Continued

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REQUIREMENTS

Contact data

Load ratings See table I

Low level 1C to 50 µA at 10 to 50 mv dc or peak ac

Intermediate current Applicable

Mixed loads Applicable

TABLE I Rated contact load and cycles (amperes per pole)

Type of load (high level)	Cycles x 10^3	28 V dc	115 V ac 1 phase 400 Hz	115/200 V ac 3 phase 400 Hz
Resistive	50	10	10	10
Inductive	10	6	N/A	N/A
Inductive	20	N/A	8	8
Motor	50	4	4	4
Lamp	50	2	2	N/A

Life 100,000 cycles, unless otherwise specified (see table I)

Contact voltage drop or resistance

Initial contact voltage drop 0 100 volt maximum Rated resistive current at 6 V dc or peak ac Relays shall not make or break this load.

Initial contact resistance 0 010 ohm maximum 50 mA max at 6 V dc or peak ac

High level life (contact voltage drop)

During life 10 percent open circuit voltage maximum

After life 0 125 volt maximum

Intermediate current (contact resistance)

During intermediate current 3 ohms maximum

After intermediate current 0 150 ohm maximum

Low level (contact resistance)

During low level 100 ohms maximum

After low level 0 150 ohm maximum at 100 mA and 28 V dc

Contact bounce 1 0 millisecond maximum

Contact stabilization time 2 5 milliseconds maximum

Overload current 30 amperes dc, 60 amperes ac

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Rupture current: 40 amperes dc, 80 amperes ac

Time current relay characteristics See table II

TABLE II Time current relay characteristics 1/

Successive application	Amperes	Time
1	20	1 hour
2	30	20 seconds
3	75	2 seconds
4	150	0 53 second

1/ All relays shall withstand overload and fault currents Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table II Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz 3-phase Cooling time between successive applications shall be 30 minutes Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Coil data 1/: See table III.

Operate time: 6 milliseconds maximum with rated coil voltage at 25°C 7 milliseconds maximum with rated coil voltage over the temperature range

Release time: 6 milliseconds maximum from rated coil voltage at 25°C. 7 milliseconds maximum from rated coil voltage over the temperature range

Duty rating: Continuous

Coil transient suppression Not applicable

Neutral screen: Not applicable

Electrical data:

Insulation resistance

Initial: 100 megohms at 500 V dc.

After life or environmental tests 50 megohms at 500 V dc

1/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations

a. Row to row assisting fields 125 (3 18 mm).

b Row to row opposing fields 1875 (4 763 mm)

c Side to side alternating fields 0625 (1 588 mm)

d. Side to side like fields 125 (3 18 mm)

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TABLE III Dash numbers and characteristics 1/

Dash number 2/			Mount	Coil data											
				At 25°C						Over temperature range			Maximum pickup voltage	
Solder pin	Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil resistance (ohms) minimum	Specified pickup voltage (V dc) 4/	Specified hold voltage (V dc) 4/	Specified dropout voltage (V dc) 4/	Specified pickup voltage (V dc) 4/	Specified hold voltage (V dc) 4/	Specified dropout voltage (V dc) 4/	High temperature test	Continuous current test
				Rated	Max									
001	002	003	No mount	6	7 3	15	3 3	1 6	0 4	4 5	2 3	0 25	5 0	5 7
004	005	006	RVFM	6	7 3	15	3 3	1 6	0 4	4 5	2 3	0 25	5 0	5 7
007	008	---	HFM	6	7 3	15	3 3	1 6	0 4	4.5	2.3	0.25	5 0	5 7
009	010	011	No mount	12	14 5	70	6 6	3 3	0 75	9 0	4 5	0 5	9 9	11 25
012	013	014	RVFM	12	14 5	70	6 6	3 3	0 75	9 0	4.5	0.5	9 9	11 25
015	016	---	HFM	12	14 5	70	6 6	3 3	0 75	9 0	4 5	0.5	9 9	11 25
017	018	019	No mount	28	29 5/	360	13 5	5 5	2 3	18 0	7 0	1 5	19 8	22 5
020	021	022	RVFM	28	29 5/	360	13 5	5 5	2.3	18 0	7 0	1 5	19 8	22 5
023	024	---	HFM	28	29 5/	360	13 5	5 5	2.3	18 0	7 0	1 5	19.8	22 5
025	026	027	No mount	48	50	1125	24 0	10 0	3 0	36 0	12 0	2 4	40 0	44 0
028	029	030	RVFM	48	50	1125	24 0	10 0	3.0	36 0	12 0	2 4	40 0	44 0
031	032	---	HFM	48	50	1125	24 0	10 0	3 0	36 0	12 0	2 4	40 0	44 0

1/ Each relay possesses high level and low level capabilities However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number Failure rate level (percent per 10,000 cycles) L, 3 0, M, 1 0; P, 0 1; R, 0 01. Example: 001L, 002R

3/ CAUTION: The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay

4/ Pickup, hold, and dropout voltages as shown are for test purposes only and are not to be used for design criteria

5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc

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Dielectric withstanding voltage 2/

Sea level	Coil to case	All other points
Initial	1,000	1,250
After life	700	1,000
Altitude		
At 80,000 feet	250	250
At 300,000 feet	500	500

Environmental characteristics

Temperature range -70°C to +125°C

Maximum altitude rating 300,000 feet

Shock (specified pulse) Applicable, MIL-STD-202, method 213, test condition C, 200 g's for 6 ±1 ms, except horizontal flange mount peak g value shall be 100 g's for 6 ±1 ms Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Vibration (sinusoidal) Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 70 to 3,000 Hz Horizontal flange mount shall be 20 g's, 57 to 3,000 Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random) Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202, method 214, test condition IG (0.4 g^2/Hz 50 to 2000 Hz), 15 minutes each plane, for horizontal flange mount test condition IE (0.2 g^2/Hz) Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Acceleration Applicable, except 15 g's.

Physical data

Dimensions and configurations See figure 1

Weight 0.062 pound (28.15 g's).

Terminal strength

Solder hook terminals

Pull force 5 pounds, ±0.5 pound

Bend: Not applicable

Twist: Not applicable.

Solder pin terminals:

Pull force:

For .030 diameter terminals: 3 pounds, ±0.3 pound

For .062 diameter terminals 10 pounds, ±1.0 pound

2/ Dielectric may be improved by suitable insulation of terminals and wiring after installation

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Bend Not applicable

Twist Not applicable

Socket pin terminals

Pull force

For .040 diameter terminals 5 pounds, ±0.5 pound

For .062 diameter terminals 10 pounds, ±0.5 pound

Bend Not applicable

Twist Not applicable

Terminal solderability Applicable to solder pin and solder hook terminals only

Seal Hermetic

Marking Applicable

Part or Identifying Number (PIN) M83536/21 (dash number from table III and suffix letter designating failure rate level) PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number

Qualification inspection:

Qualification inspection and sample size See table IV

TABLE IV Qualification inspection and sample size 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/	M83536/21-022	18 units plus 1 open unit for level L at C = 0 2/
33 units plus 1 open unit for level M at C = 0 2/	"	33 units plus 1 open unit for level M at C = 0 2/
Qualification inspection as applicable	"	Qualification inspection as applicable
	M83536/21-018 M83536/21-023	2 units, qualification inspection table, group II, shock, vibration, acceleration, terminal strength, and seal
	M83536/21-002 M83536/21-014 M83536/21-031	2 units, qualification inspection table, group II

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-83536/22 may be used in addition to MIL-R-83536/21 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II of MIL-R-83536, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

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Qualification inspection (reduced testing) (sample size - 2 units each coil voltage and 1 unsealed unit). See table V

If the relays produced for MIL-R-83536/21 are similar in construction and design except for the suppression network to the relays produced for MIL-R-83536/22 then reduced testing for qualification of MIL-R-83536/21 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-83536/22 relays. For reduced testing see table V

TABLE V Qualification inspection (reduced testing)

Inspection
2 units each coil voltage Group II of qualification inspection table 1 unsealed sample unit for internal inspection

Supersession data See table VI

TABLE VI Supersession data

Superseded PIN M6106/29-	Replacement PIN M83536/21-
001	016
002	016
009	017
010	017
013	018
014	018
017	019
018	019
021	020
022	020
025	021
026	021
029	022
030	022

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

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